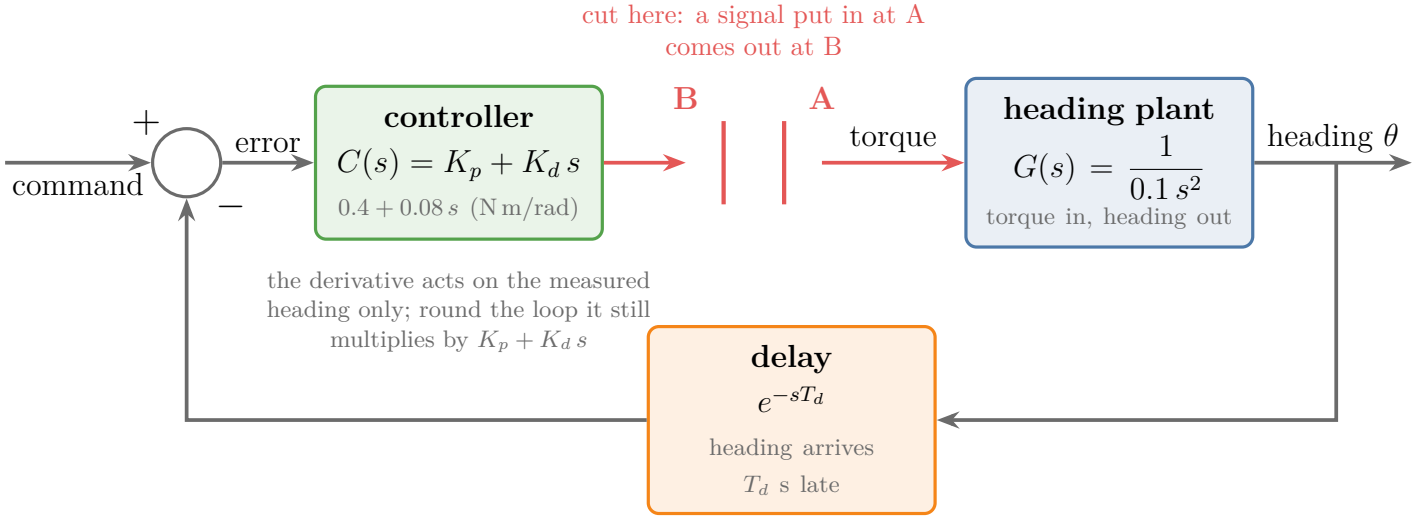

cut here: a signal put in at A
comes out at B
command
+
−
error
controller
$C(s) = K_p + K_d\,s$
$0.4 + 0.08\,s$ (N m/rad)
B
A
torque
heading plant
$G(s) = \dfrac{1}{0.1\,s^2}$
torque in, heading out
heading θ
the derivative acts on the measured heading only; round the loop it still multiplies by $K_p + K_d\,s$
delay
e^{-sT_d}
heading arrives
T_d s late
once round, A to B: $C(s)\,G(s)\,e^{-sT_d}$ and the minus sign,
so B $= -L(s)\times$ A with the loop transfer function $L(s) = C(s)\,G(s)\,e^{-sT_d}$